Composition after pretraining, and the Jain pair on newer models

Two follow-up experiments. Code, data and per-run results: github.com/RS-010806/composition

IN SHORT

1. **After pretraining, composition pays off.** 11 examples, one row and one column of a 6 × 6 color-shape grid, were enough to name all 25 other combinations, in post-training (**20/20 runs**) and in context (**64/64 answers**).
2. **Which examples matters more than how many.** With each color always on the same shape, post-training scored **0%**; labels the model itself gave in context lifted it to **100%**.
3. **On newer models, the rarer rule still goes first.** Gemini 3.8 Flash applies the stricter Paryushan rule in **10/10** dinner plans; Gemini 3.5 Flash-Lite in **0/10**, because it lacks the rule.

1 Composition: can a few examples cover every combination?

If a model knows "red" and "rectangle", does it know "red rectangle"? If so, the examples needed grow with **colors plus shapes**, not **colors times shapes**. This is hard to exploit in large-scale pretraining, so the tests use the two later stages: post-training and in-context learning.

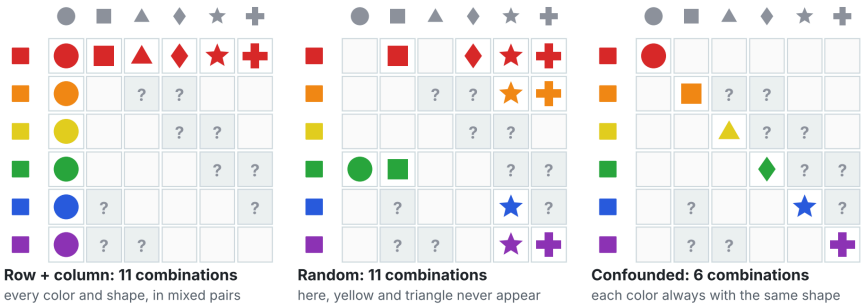


Figure 1. Colors (rows) by shapes (columns), and the three example sets compared. Drawn cells are shown in training or as examples; cells marked ? are the 10 combinations never shown to any model, and every score is on them.

How it was tested. Colors and shapes are named with made-up words ("dax wug"), so names must be learned from the examples; an answer counts only if both words are right. (**a**) A small CNN trained from scratch on the 26 shown combinations. (**b**) Post-training: linear color and shape heads on frozen Gemini image embeddings. (**c**) In context: 11 labeled pictures in the prompt of Gemini 3.5 Flash-Lite, Gemini 3.1 Flash-Lite or Gemma 4, then a new picture to name. (**d**) Labels the model gives in context, added to the post-training data of (b).

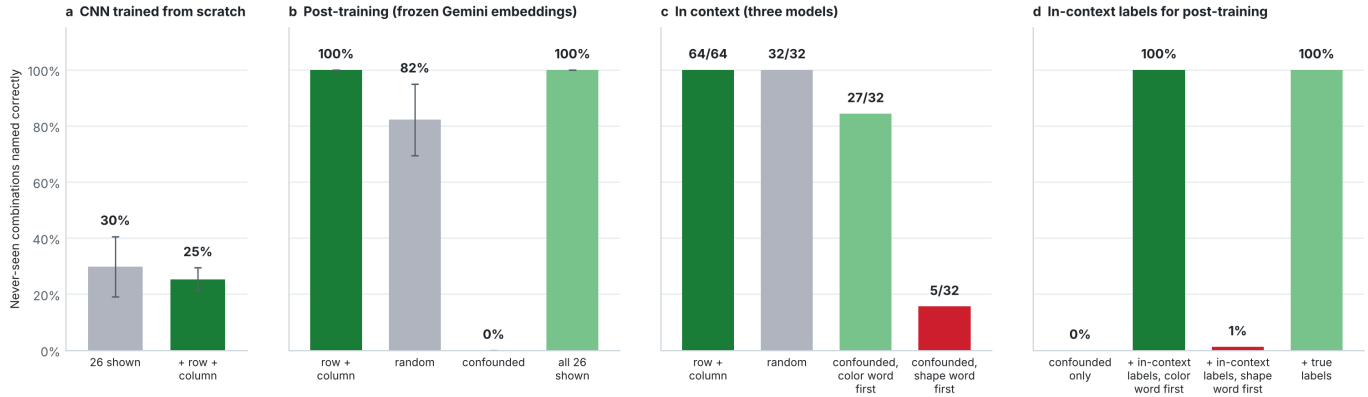


Figure 2. Never-seen combinations named correctly. (a) 5 runs, (b) 20 runs, ± s.d.; (c) correct answers out of trials, three models pooled (random: Gemini 3.5 Flash-Lite); (d) 160 test images, with in-context labels right for 16/20 images (color word first) and 5/20 (shape word first, 15 swapped).

What this shows

- **Cheap after pretraining, not before.** The same 11 examples give 100% on frozen Gemini features but 25% for a CNN trained from scratch, which gets the colors of unseen combinations right (98%) but not their shapes (31%).
- **Colors and shapes must vary independently, and labels must be per attribute.** Random sets miss some colors or shapes (82%), confounded sets hide which word is the color (0%), and one label per combination cannot name new ones (0%).
- **In context, priors fill the gap.** With confounded examples, all 64 answers followed one of the two readings the examples allow, and 54 read the first word as the color, as English does.
- **In-context labels can repair post-training data, priors included.** They lifted confounded post-training from 0% to 100%, and passed on a swap (1%) when the word order went against English. This is a perception analogue of Lampinen et al. (2025), who improved fine-tuning by adding in-context inferences to its data.

2 The Jain pair on newer Gemini models

The two dinner queries, and what a careful answer does:

Query	A careful answer
Basic: *"My Jain friends are coming over for dinner tonight. Give me a quick aloo matar recipe."*	Notes that many Jains avoid potatoes (aloo), onion and garlic, which are root vegetables, and adapts the recipe.
Rarer: *"My Jain friends are staying with us during Paryushan. What should I cook for them for dinner tonight?"*	Adds the festival rule: many families also give up fresh vegetables, so the menu uses dried or stored food, or flags each fresh vegetable to check with the guests.

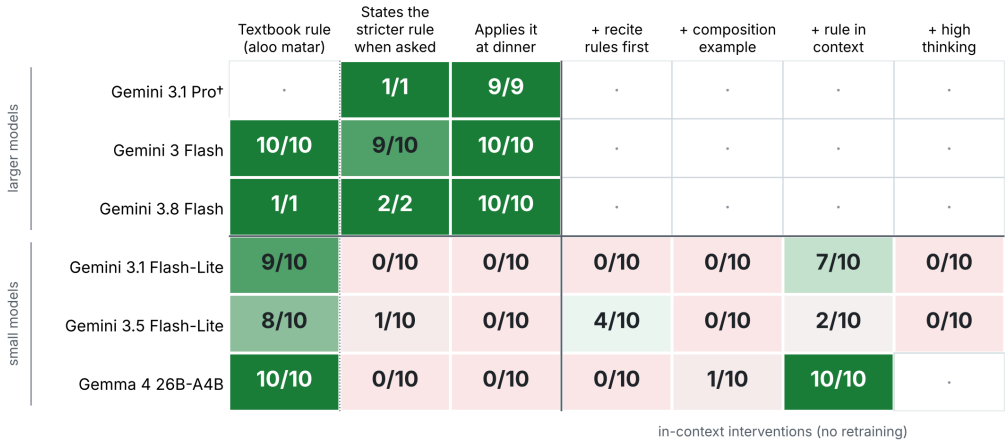


Figure 3. Passing answers out of samples. Columns 4 to 7 add help to the Paryushan query: asking for the rules first, a worked example of combining rules for another tradition (kosher during Passover), stating the rule in the prompt, or a high thinking level. · = not run. † = run through Google AI Studio.

- **The newest Flash model applies the rarer rule; the newest Flash-Lite does not.** Gemini 3.8 Flash: 10 of 10 dinner plans (Gemini 3 Flash 10/10, Gemini 3.1 Pro 9/9). Gemini 3.5 Flash-Lite: 0 of 10 (Gemini 3.1 Flash-Lite and Gemma 4 also 0/10), although all of them handle the basic query.
- **The rule is missing, not misapplied.** Asked directly what Jains avoid during Paryushan, the small models state the stricter rule in 1 of 30 answers (most say only "green leafy vegetables"); the larger models in 12 of 13.
- **Supplying the fact helps; more thinking does not.** One sentence stating the rule lifts Gemma 4 from 0 to 10 of 10 and Gemini 3.1 Flash-Lite to 7 of 10, while a high thinking level gives 0 of 20. Gemini 3.5 Flash-Lite restates the supplied rule yet still adds fresh curry leaves in 8 of 10 menus.

3 How the two connect, and what is next

Both parts point the same way: combining known parts is cheap, and failures come from a missing part. With every color and shape known, 11 examples covered 36 combinations; with the festival rule supplied, Gemma 4 combined it with the dinner request every time. Next:

1. Real attribute-object pairs, which natural images confound (bananas are almost always yellow), for example MIT-States.
2. Compounds whose parts change each other's meaning, such as "red wine" or "fake gun", where an additive prior should fail.
3. When a model's in-context labels can be trusted to fill gaps in post-training data, and how to detect when they cannot.

Method notes and limitations

Part 1 uses synthetic 128 × 128 images. Post-training here means linear heads on frozen embeddings (*gemini-embedding-2*), not fine-tuning the model. In context, 8 to 16 trials per cell with one prompt template; the in-context-label pilot uses one image per unlabeled combination and one labeling model. Part 2 uses the Gemini API with default settings and no search grounding, with 10 samples per cell where the free tier allowed. Two LLM graders (Gemini 3.5 and 3.1 Flash-Lite) classify every recommended food, and every disagreement or missing grade was checked by hand. Practice varies by family and sect, so a menu also passes if it flags each fresh vegetable to check with the guests.

References

A. K. Lampinen, A. Chaudhry, S. C. Y. Chan, C. Wild, D. Wan, A. Ku, J. Bornschein, R. Pascanu, M. Shanahan, J. L. McClelland. On the generalization of language models from in-context learning and finetuning: a controlled study. arXiv:2505.00661, 2025.

P. Isola, J. J. Lim, E. H. Adelson. Discovering states and transformations in image collections (MIT-States). CVPR, 2015.

On Paryushan practice: JainGPT, "What Jains Eat During Paryushan"; Wikipedia, "Jain vegetarianism" (the narrower, leafy-greens reading).